

Computer Vision Session 1

User Manual

1. Introduction

This project contains two interactive Computer Vision applications developed using Python and OpenCV:

- **Mini Painter:** An interactive drawing application for creating circles, rectangles, and polygons using the mouse and keyboard.
 - **Object Counter:** A color-based object detection application that detects and counts objects of selected colors in an image.
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2. Requirements

The following are required to run the project:

- Python 3.x
- OpenCV
- NumPy
- A computer with a keyboard and mouse

The required Python libraries can be installed using:

```
pip install -r requirements.txt
```

3. Mini Painter

3.1 Description

Mini Painter is an interactive drawing application that allows the user to draw different shapes on a canvas.

The user can:

- Select a shape.
- Select a drawing color.
- Draw the selected shape using the mouse.
- View the currently selected shape and color.
- Save the drawing as an image.

3.2 How to Run

Open a terminal in the `miniPainter` directory and run:

```
python main.py
```

A window named **Mini Painter** will appear.

3.3 Controls

Key Action

- C Select Circle
- R Select Rectangle
- P Select Polygon
- 1 Select White
- 2 Select Red
- 3 Select Green
- 4 Select Blue
- 5 Select Yellow
- W Save the drawing
- Q Quit

3.4 How to Use

1. Run the application.
2. Select a shape using C, R, or P.
3. Select a color using keys 1 to 5.
4. Click anywhere on the canvas using the left mouse button.
5. The selected shape will be drawn at the clicked location.
6. The current shape and color are displayed in the upper-left corner of the window.
7. Continue selecting different shapes and colors as needed.

3.5 Saving the Drawing

Press:

W

The current canvas will be saved as:

mini_painter_output.png

3.6 Exiting

Press:

Q

to close the application.

4. Object Counter

4.1 Description

Object Counter is an image-processing application that detects and counts objects based on their colors.

The application supports:

- Red objects
- Green objects
- Blue objects

The application processes the provided test image and draws bounding boxes around detected objects.

4.2 How to Run

Open a terminal in the project root directory and run:

```
python -m object_counter.main
```

The application will open the test image located at:

```
test_images/image.png
```

4.3 Controls

Key Action

- 1 Detect Red objects
- 2 Detect Green objects
- 3 Detect Blue objects
- Q Quit

4.4 How to Use

1. Run the application.
2. The test image will appear.
3. Press 1 to detect red objects.
4. Press 2 to detect green objects.
5. Press 3 to detect blue objects.
6. The detected objects will be surrounded by bounding boxes.
7. The selected color and total number of detected objects will be displayed on the image.
8. The detected object count will also be printed in the terminal.
9. Select another color without restarting the application.

4.5 Output

For each selected color, the application displays:

- The selected color.
- The number of detected objects.
- Bounding boxes around the detected objects.

The terminal also prints the detected count.

Example:

Red objects: 3

Green objects: 2

Blue objects: 4

The exact counts depend on the objects present in the test image.

4.6 Exiting

Press:

Q

to close the application.

5. Troubleshooting

Mini Painter does not start

Make sure that:

- Python is installed.
- OpenCV and NumPy are installed.
- The terminal is opened inside the miniPainter directory.
- The command is written correctly:

```
python main.py
```

Object Counter does not start

Make sure that:

- Python is installed.
- OpenCV and NumPy are installed.
- The terminal is opened in the project root directory.
- The test image exists at:

```
test_images/image.png
```

- The application is started using:

```
python -m object_counter.main
```

No objects are detected

Make sure that:

- The selected color exists in the image.
- The objects have sufficiently distinct solid colors.
- The correct color key is being used.

6. Project Structure

The project is organized into separate components for each task:

CV_Session1/

```
|
|— miniPainter/
|
|— object_counter/
|
|— test_images/
|
|— docs/
|
|— README.md
|— requirements.txt
|— .gitignore
```

The source code is separated into classes according to their responsibilities to keep the project organized and maintainable.